

# NATA SUPERMARKET



GROUP 6

BUSINESS ANALYTICS

# OBJECTIVE

TO UNDERSTAND THE DATA GATHERED  
TO MASTER THE FORECASTING OF  
FUTURE SALES



VINA VERAGO

## INITIAL PROBLEM

2021

Nata Supermarket had low interaction with data analysis, and they noticed that not only their competitors but also their stakeholders were experts in forecasting

- **PRODUCT INVENTORY MANAGEMENT**
- **PREDICT SHOPPING HABITS BY GROUP**



CLEAN → CLUSTER → FORECAST → INVERNTORY ANALYSIS

# DATA CLEANING

## Data: 2,240 rows

- 24 missing values in “Income”.
- 3 unknown columns.
- No duplicated data.



## Data: 2,240 rows

- Replace the missing values with mode.
- Drop those unknown columns.

RangeIndex: 2240 entries, 0 to 2239

Data columns (total 29 columns):

| #  | Column              | Non-Null Count | Dtype   |
|----|---------------------|----------------|---------|
| 0  | ID                  | 2240 non-null  | int64   |
| 1  | Year_Birth          | 2240 non-null  | int64   |
| 2  | Education           | 2240 non-null  | object  |
| 3  | Marital_Status      | 2240 non-null  | object  |
| 4  | Income              | 2216 non-null  | float64 |
| 5  | Kidhome             | 2240 non-null  | int64   |
| 6  | Teenhome            | 2240 non-null  | int64   |
| 7  | Dt_Customer         | 2240 non-null  | object  |
| 8  | Recency             | 2240 non-null  | int64   |
| 9  | MntWines            | 2240 non-null  | int64   |
| 10 | MntFruits           | 2240 non-null  | int64   |
| 11 | MntMeatProducts     | 2240 non-null  | int64   |
| 12 | MntFishProducts     | 2240 non-null  | int64   |
| 13 | MntSweetProducts    | 2240 non-null  | int64   |
| 14 | MntGoldProds        | 2240 non-null  | int64   |
| 15 | NumDealsPurchases   | 2240 non-null  | int64   |
| 16 | NumWebPurchases     | 2240 non-null  | int64   |
| 17 | NumCatalogPurchases | 2240 non-null  | int64   |
| 18 | NumStorePurchases   | 2240 non-null  | int64   |
| 19 | NumWebVisitsMonth   | 2240 non-null  | int64   |
| 20 | AcceptedCmp3        | 2240 non-null  | int64   |
| 21 | AcceptedCmp4        | 2240 non-null  | int64   |
| 22 | AcceptedCmp5        | 2240 non-null  | int64   |
| 23 | AcceptedCmp1        | 2240 non-null  | int64   |
| 24 | AcceptedCmp2        | 2240 non-null  | int64   |
| 25 | Complain            | 2240 non-null  | int64   |
| 26 | Z_CostContact       | 2240 non-null  | int64   |
| 27 | Z_Revenue           | 2240 non-null  | int64   |
| 28 | Response            | 2240 non-null  | int64   |

# OUTLINER

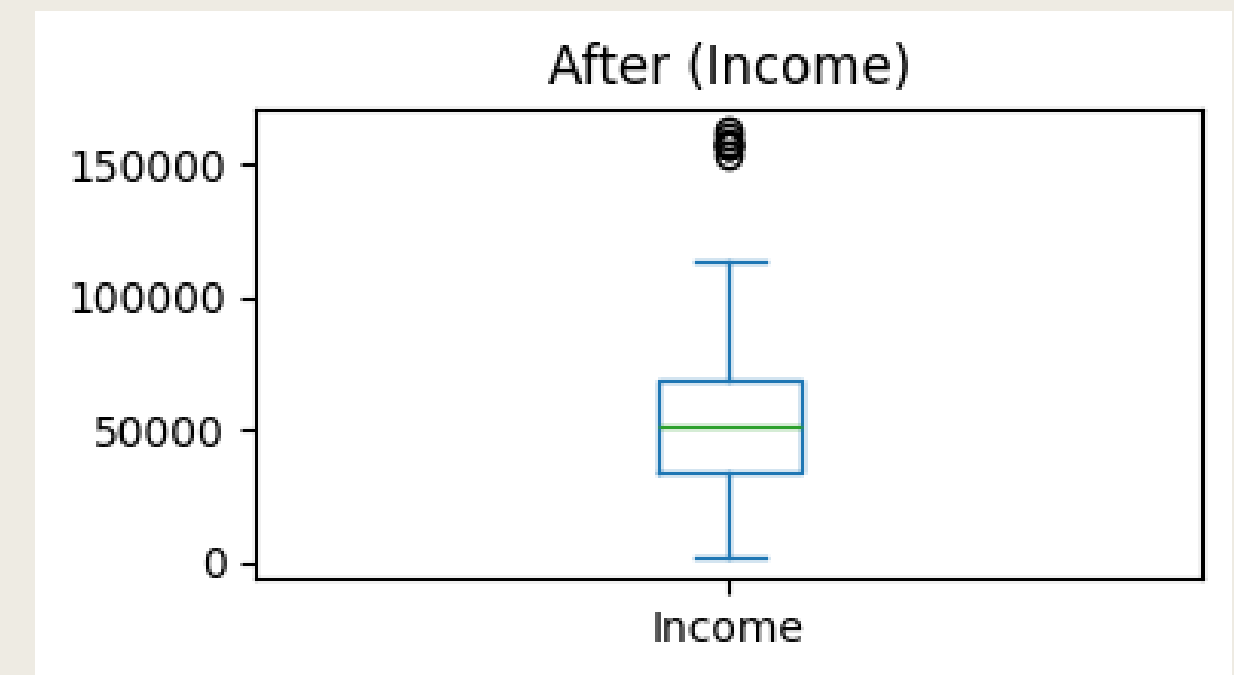
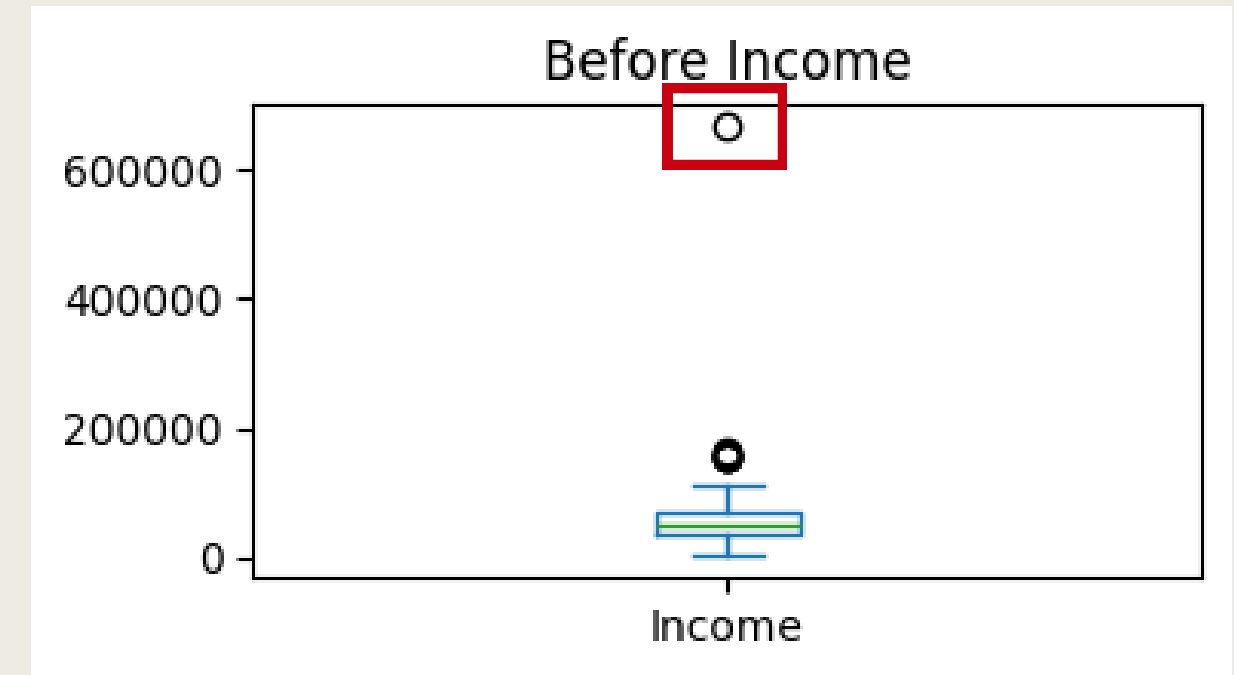
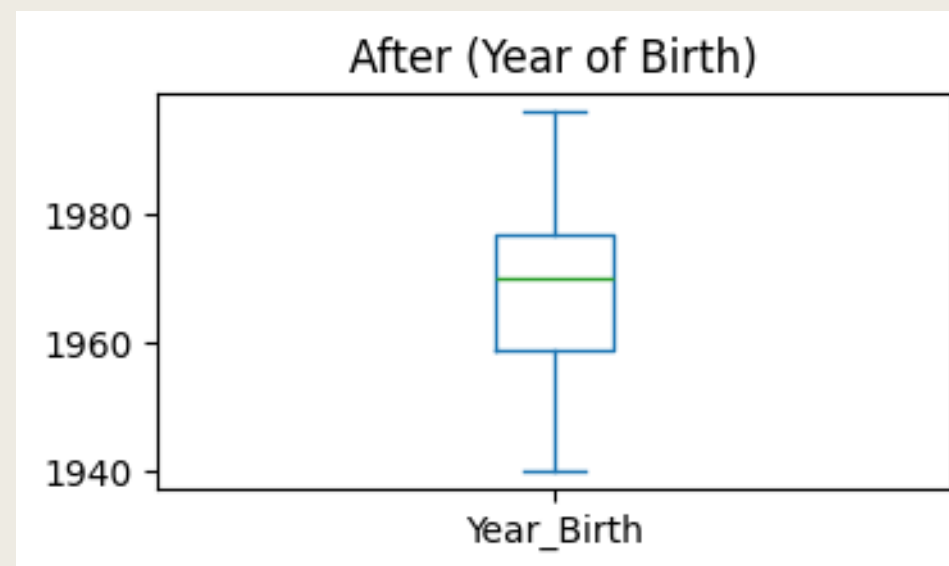
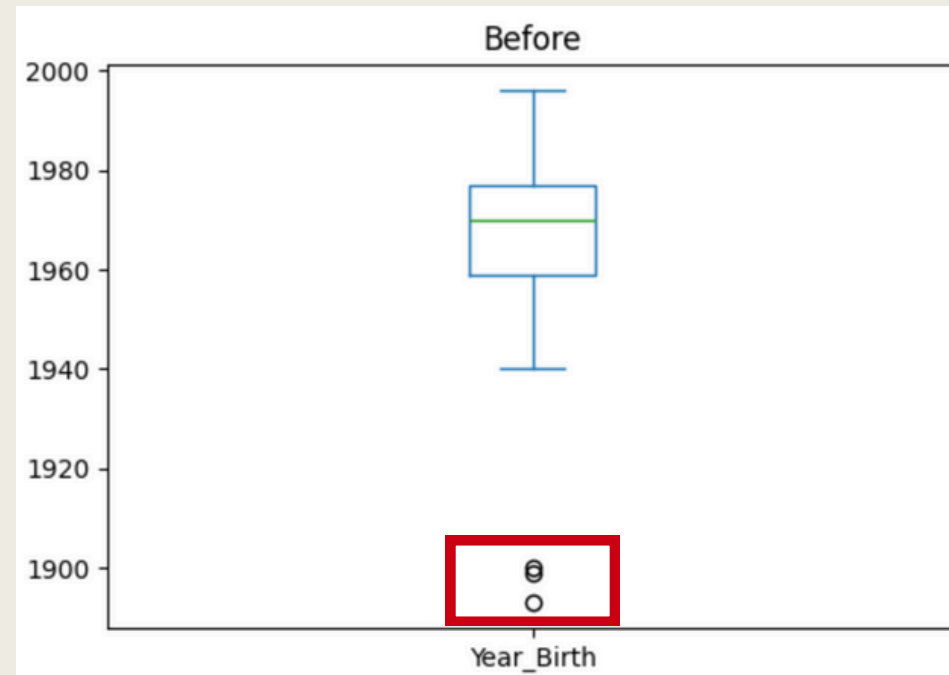
**Data: 2,240 rows**

- 3 outliers in "Year\_Birth".
- 1 outlier in "Income".



**Data: 2,236 rows**

- Drop those outliers



# BEFORE CLUSTERING...

**Data: 2,236 rows**

- Encoding "Education"
- Encoding "Marital\_Status"
- Weird values Absurd & YOLO



**Data: 2,232 rows**

- Drop those weird values

| Education  |      |
|------------|------|
| Graduation | 1126 |
| PhD        | 485  |
| Master     | 370  |
| 2n Cycle   | 201  |
| Basic      | 54   |



- Basic = 0
- Graduation = 1
- Master = 2
- 2n Cycle = 3
- PhD = 4

| Marital_Status |     |
|----------------|-----|
| Married        | 864 |
| Together       | 578 |
| Single         | 479 |
| Divorced       | 231 |
| Widow          | 77  |
| Alone          | 3   |
| Absurd         | 2   |
| YOLO           | 2   |



Married = 0  
Together = 0  
Single = 1  
Divorced = 1  
Widow = 1  
Alone = 1

# CLUSTERING

```
Index: 2232 entries, 0 to 2235
Data columns (total 26 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                     2232 non-null   int64
1   Year_Birth                           2232 non-null   int64
2   Education                             2232 non-null   int64
3   Marital_Status                       2232 non-null   int64
4   Income                               2232 non-null   int64
5   Kidhome                              2232 non-null   int64
6   Teenhome                             2232 non-null   int64
7   Dt_Customer                          2232 non-null   datetime64[ns]
8   Recency                              2232 non-null   int64
9   MntWines                             2232 non-null   int64
10  MntFruits                             2232 non-null   int64
11  MntMeatProducts                       2232 non-null   int64
12  MntFishProducts                       2232 non-null   int64
13  MntSweetProducts                     2232 non-null   int64
14  MntGoldProds                          2232 non-null   int64
15  NumDealsPurchases                     2232 non-null   int64
16  NumWebPurchases                       2232 non-null   int64
17  NumCatalogPurchases                  2232 non-null   int64
18  NumStorePurchases                    2232 non-null   int64
19  NumWebVisitsMonth                    2232 non-null   int64
20  AcceptedCmp3                         2232 non-null   int64
21  AcceptedCmp4                         2232 non-null   int64
22  AcceptedCmp5                         2232 non-null   int64
23  AcceptedCmp1                         2232 non-null   int64
24  AcceptedCmp2                         2232 non-null   int64
25  Complain                             2232 non-null   int64
```

## Goal

- Segment customers based on all the features.
- Identify patterns for marketing strategy and customer insights.

## Approach

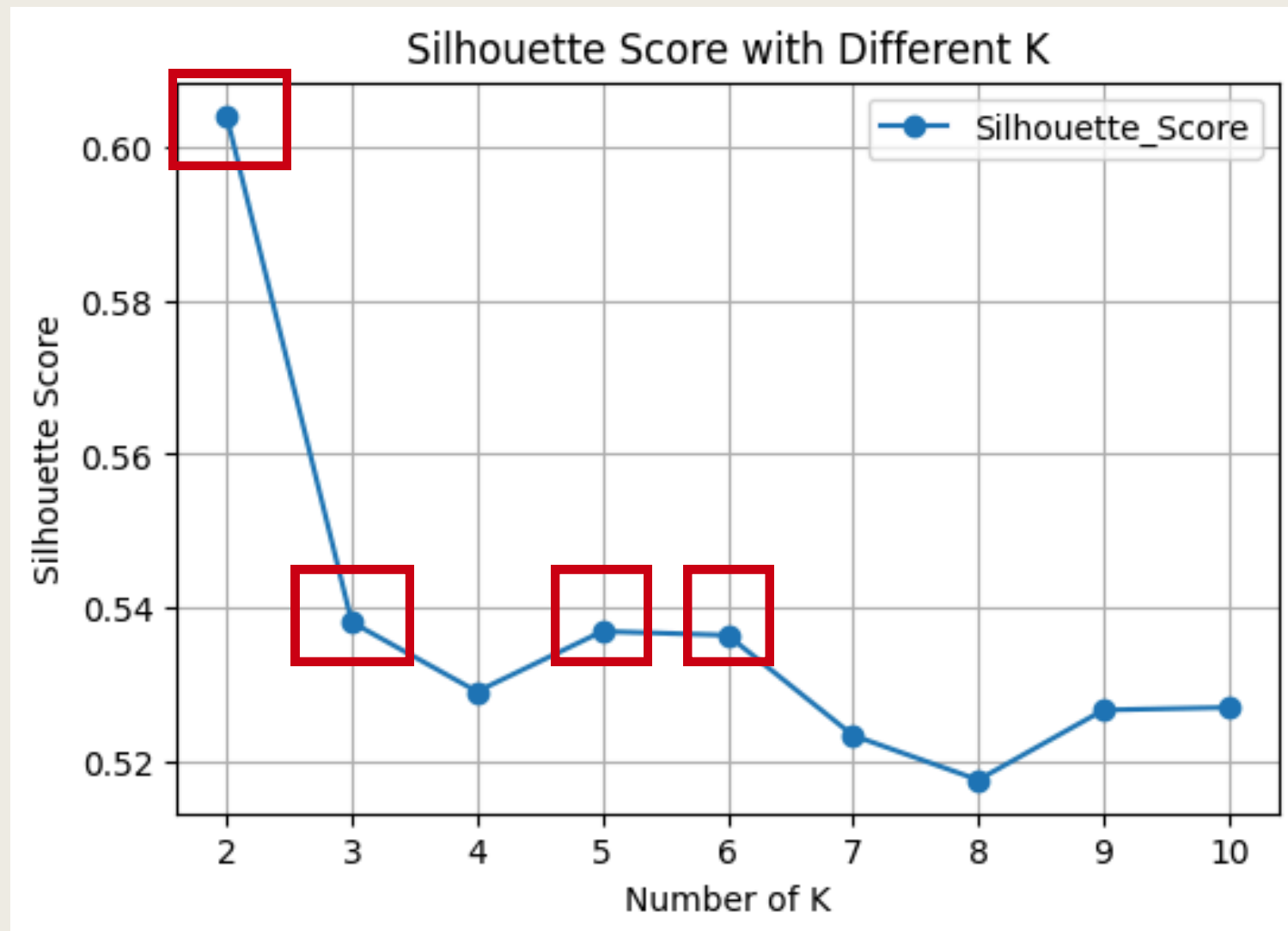
- Removed ID to avoid distortion.
- Ran K-Means with K = 2 to 10.
- Evaluated each model using Silhouette Score.

## Silhouette Score

- How well the data points fit their assigned clusters.
- Are points close to others in the same cluster?
- Are they far away from points in other clusters?

# K-MEANS CLUSTERING

- **Closer to 1**: The clusters are well separated, and each point fits its cluster nicely.
- **Around 0**: The clusters overlap; points are not clearly separated.
- **Closer to -1**: Many points may be in the wrong cluster.



## K = 2 TOO GENERIC

- Segmentation becomes too broad.
- can't clearly differentiate distinct customer groups.



## K = 5, 6 TOO FRAGMENTED

- Segments become too small.
- Hard to generalize customer needs.



## K = 3 (CHOSEN K)

- Provides a clear and balanced segmentation.

# K-MEANS CLUSTERING

**36% (801)**

## **CLUSTER 0**

mid-income customers (40,000–60,000) with moderate spending on wine, fruits, meat, fish, sweets, and gold over the past two years, with gold spending close to Cluster 2. Many have one teenager at home.

**33% (737)**

## **CLUSTER 1**

low-income customers (0–40,000) who show the lowest spending across all categories. Many have one young child. They make fewer online purchases, but recorded the highest number of website visits last month.

**31% (694)**

## **CLUSTER 2**

high-income customers (65,000–160,000) with the highest spending on wine, fruits, meat, fish, and sweets, while their gold spending is similar to Cluster 1. Most have no children.

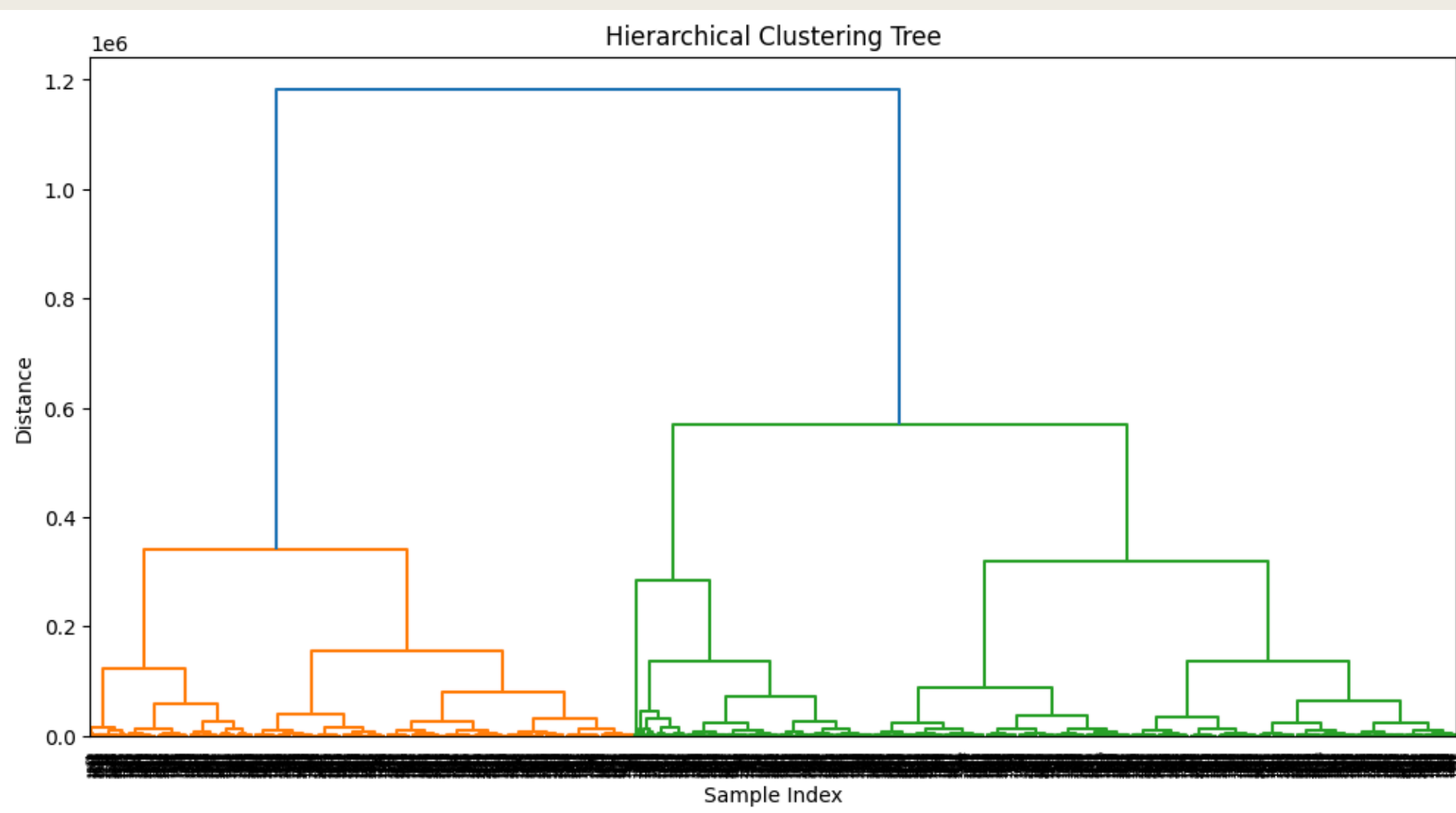
# HIERARCHICAL CLUSTERING

## Approach

- Model calculates the distance between customers
- Merges the closest customers step by step
- After the tree was built, split it into exactly 3 groups

## Found

- Distribution was 43% / 40% / 17%
- The segmentation was not balanced



**43% (953)**

**CLUSTER 0**

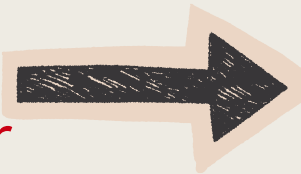
**40% (890)**

**CLUSTER 1**

**17% (389)**

**CLUSTER 2**

# FORECASTING

(03) To plan what we will buy from suppliers from the information given for each cluster  To have the correct inventory levels

Forecast for Cluster: 0, Product: Fish generated. Best AIC: 1605.96  
Forecast for Cluster: 0, Product: Fruits generated. Best AIC: 1539.82  
Forecast for Cluster: 0, Product: Gold generated. Best AIC: 1778.01  
Forecast for Cluster: 0, Product: Meat generated. Best AIC: 2016.33  
Forecast for Cluster: 0, Product: Wines generated. Best AIC: 2353.42  
Forecast for Cluster: 1, Product: Fish generated. Best AIC: 1330.49  
Forecast for Cluster: 1, Product: Fruits generated. Best AIC: 1146.45  
Forecast for Cluster: 1, Product: Gold generated. Best AIC: 1507.57  
Forecast for Cluster: 1, Product: Meat generated. Best AIC: 1688.14  
Forecast for Cluster: 1, Product: Wines generated. Best AIC: 1680.14  
Forecast for Cluster: 2, Product: Fish generated. Best AIC: 1913.56  
Forecast for Cluster: 2, Product: Fruits generated. Best AIC: 1820.79  
Forecast for Cluster: 2, Product: Gold generated. Best AIC: 1883.37  
Forecast for Cluster: 2, Product: Meat generated. Best AIC: 2424.20  
Forecast for Cluster: 2, Product: Wines generated. Best AIC: 2539.40

|                           |         |            |            |             |              |
|---------------------------|---------|------------|------------|-------------|--------------|
| Sample of 2015 Forecasts: |         |            |            |             |              |
| Date                      |         | 2014-12-14 | 2014-12-21 | 2014-12-28  | 2015-01-04 \ |
| Cluster_group             | Product |            |            |             |              |
| 0                         | Fish    | -11.424367 | -11.419361 | 149.443198  | 252.854535   |
|                           | Fruits  | -2.904624  | -2.942973  | 191.813147  | 274.871930   |
|                           | Gold    | -12.128624 | -12.084726 | 276.485536  | 571.913969   |
|                           | Meat    | 7.842546   | 7.878438   | 977.243665  | 1473.771778  |
|                           | Wines   | 43.296266  | 44.048664  | 2223.877191 | 4594.101504  |
| Date                      |         | 2015-01-11 | 2015-01-18 | 2015-01-25  | 2015-02-01 \ |
| Cluster_group             | Product |            |            |             |              |
| 0                         | Fish    | -17.489216 | -17.475684 | -17.461196  | 546.549041   |
|                           | Fruits  | -5.282427  | -5.274665  | -5.269092   | 287.105175   |
|                           | Gold    | -20.403321 | -20.330103 | -20.275031  | 867.798102   |
|                           | Meat    | 3.155520   | 3.124985   | 3.085361    | 1762.602017  |
|                           | Wines   | 44.081610  | 44.795918  | 42.311801   | 6170.282924  |

First we performed  
a weekly forecast

|               |         |            |            |             |              |              |
|---------------|---------|------------|------------|-------------|--------------|--------------|
| Date          |         | 2015-02-08 | 2015-02-15 | ...         | 2015-10-04   | 2015-10-11 \ |
| Cluster_group | Product |            |            | ...         |              |              |
| 0             | Fish    | -17.417117 | -17.401194 | ...         | 626.592797   | -17.086042   |
|               | Fruits  | -5.247909  | -5.237185  | ...         | 384.793845   | -4.884532    |
|               | Gold    | -20.155577 | -20.092592 | ...         | 1135.684856  | -19.007755   |
|               | Meat    | 3.204757   | 3.404792   | ...         | 2437.589135  | 5.465515     |
|               | Wines   | 45.511383  | 46.296691  | ...         | 5485.967508  | 66.297971    |
| Date          |         | 2015-10-18 | 2015-10-25 | 2015-11-01  | 2015-11-08 \ |              |
| Cluster_group | Product |            |            |             |              |              |
| 0             | Fish    | -17.089013 | -17.103612 | 542.554356  | -17.126227   |              |
|               | Fruits  | -4.877636  | -4.866836  | 409.831365  | -4.850344    |              |
|               | Gold    | -19.065835 | -19.091319 | 865.196888  | -19.199092   |              |
|               | Meat    | 5.461161   | 5.435809   | 1616.920756 | 5.693222     |              |
|               | Wines   | 67.009622  | 64.599425  | 5589.104295 | 64.540568    |              |
| Date          |         | 2015-11-15 | 2015-11-22 | 2015-11-29  | 2015-12-06   |              |
| Cluster_group | Product |            |            |             |              |              |
| 0             | Fish    | -17.135817 | -17.161536 | 434.497506  | 195.152104   |              |
|               | Fruits  | -4.840393  | -4.833451  | 389.761271  | 162.943657   |              |
|               | Gold    | -19.268725 | -19.321965 | 648.706043  | 93.289687    |              |
|               | Meat    | 5.736326   | 5.854957   | 1105.783662 | 742.655820   |              |
|               | Wines   | 65.289458  | 66.040985  | 3545.125973 | 2647.240313  |              |

[5 rows x 52 columns]

But the quantities required for every product were not enough to perform a purchase, and for the first weeks, the requirement was 0:

| Sample of 2015 Forecasts after setting negative values to zero: |         |            |            |             |             |              |
|-----------------------------------------------------------------|---------|------------|------------|-------------|-------------|--------------|
| Date                                                            |         | 2014-12-14 | 2014-12-21 | 2014-12-28  | 2015-01-04  | \            |
| Cluster_group                                                   | Product |            |            |             |             |              |
| 0                                                               | Fish    | 0.000000   | 0.000000   | 149.443198  | 252.854535  |              |
|                                                                 | Fruits  | 0.000000   | 0.000000   | 191.813147  | 274.871930  |              |
|                                                                 | Gold    | 0.000000   | 0.000000   | 276.485536  | 571.913969  |              |
|                                                                 | Meat    | 7.842546   | 7.878438   | 977.243665  | 1473.771778 |              |
|                                                                 | Wines   | 43.296266  | 44.048664  | 2223.877191 | 4594.101504 |              |
|                                                                 |         |            |            |             |             |              |
| Date                                                            |         | 2015-01-11 | 2015-01-18 | 2015-01-25  | 2015-02-01  | \            |
| Cluster_group                                                   | Product |            |            |             |             |              |
| 0                                                               | Fish    | 0.000000   | 0.000000   | 0.000000    | 546.549041  |              |
|                                                                 | Fruits  | 0.000000   | 0.000000   | 0.000000    | 287.105175  |              |
|                                                                 | Gold    | 0.000000   | 0.000000   | 0.000000    | 867.798102  |              |
|                                                                 | Meat    | 3.15552    | 3.124985   | 3.085361    | 1762.602017 |              |
|                                                                 | Wines   | 44.08161   | 44.795918  | 42.311801   | 6170.282924 |              |
|                                                                 |         |            |            |             |             |              |
| Date                                                            |         | 2015-02-08 | 2015-02-15 | ...         | 2015-10-04  | 2015-10-11 \ |
| Cluster_group                                                   | Product |            |            | ...         |             |              |
| 0                                                               | Fish    | 0.000000   | 0.000000   | ...         | 626.592797  | 0.000000     |
|                                                                 | Fruits  | 0.000000   | 0.000000   | ...         | 384.793845  | 0.000000     |
|                                                                 | Gold    | 0.000000   | 0.000000   | ...         | 1135.684856 | 0.000000     |
|                                                                 | Meat    | 3.204757   | 3.404792   | ...         | 2437.589135 | 5.465515     |
|                                                                 | Wines   | 45.511383  | 46.296691  | ...         | 5485.967508 | 66.297971    |

So we preferred a  
monthly forecast...

### Sample of 2015 DES Forecasts:

| Date          |         | 2015-01-01  | 2015-02-01  | 2015-03-01  | 2015-04-01  | \ |
|---------------|---------|-------------|-------------|-------------|-------------|---|
| Cluster_group | Product |             |             |             |             |   |
| 0             | Fish    | 519.528586  | 518.319265  | 517.112760  | 515.909063  |   |
|               | Fruits  | 398.902202  | 398.524661  | 398.147478  | 397.770651  |   |
|               | Gold    | 937.464417  | 934.386151  | 931.317992  | 928.259909  |   |
|               | Meat    | 2103.992004 | 2102.233177 | 2100.474351 | 2098.715525 |   |
|               | Wines   | 5983.315339 | 5969.011701 | 5954.708062 | 5940.404424 |   |

| Date          |         | 2015-05-01  | 2015-06-01  | 2015-07-01  | 2015-08-01  | \ |
|---------------|---------|-------------|-------------|-------------|-------------|---|
| Cluster_group | Product |             |             |             |             |   |
| 0             | Fish    | 514.708168  | 513.510068  | 512.314757  | 511.122228  |   |
|               | Fruits  | 397.394182  | 397.018068  | 396.642311  | 396.266909  |   |
|               | Gold    | 925.211866  | 922.173832  | 919.145774  | 916.127659  |   |
|               | Meat    | 2096.956699 | 2095.197873 | 2093.439047 | 2091.680221 |   |
|               | Wines   | 5926.100786 | 5911.797147 | 5897.493509 | 5883.189871 |   |

| Date          |         | 2015-09-01  | 2015-10-01  | 2015-11-01  | 2015-12-01  |  |
|---------------|---------|-------------|-------------|-------------|-------------|--|
| Cluster_group | Product |             |             |             |             |  |
| 0             | Fish    | 509.932476  | 508.745492  | 507.561272  | 506.379808  |  |
|               | Fruits  | 395.891863  | 395.517171  | 395.142834  | 394.768852  |  |
|               | Gold    | 913.119455  | 910.121127  | 907.132646  | 904.153977  |  |
|               | Meat    | 2089.921395 | 2088.162569 | 2086.403743 | 2084.644916 |  |
|               | Wines   | 5868.886233 | 5854.582594 | 5840.278956 | 5825.975318 |  |

## Double Exponential Smoothing Forecast

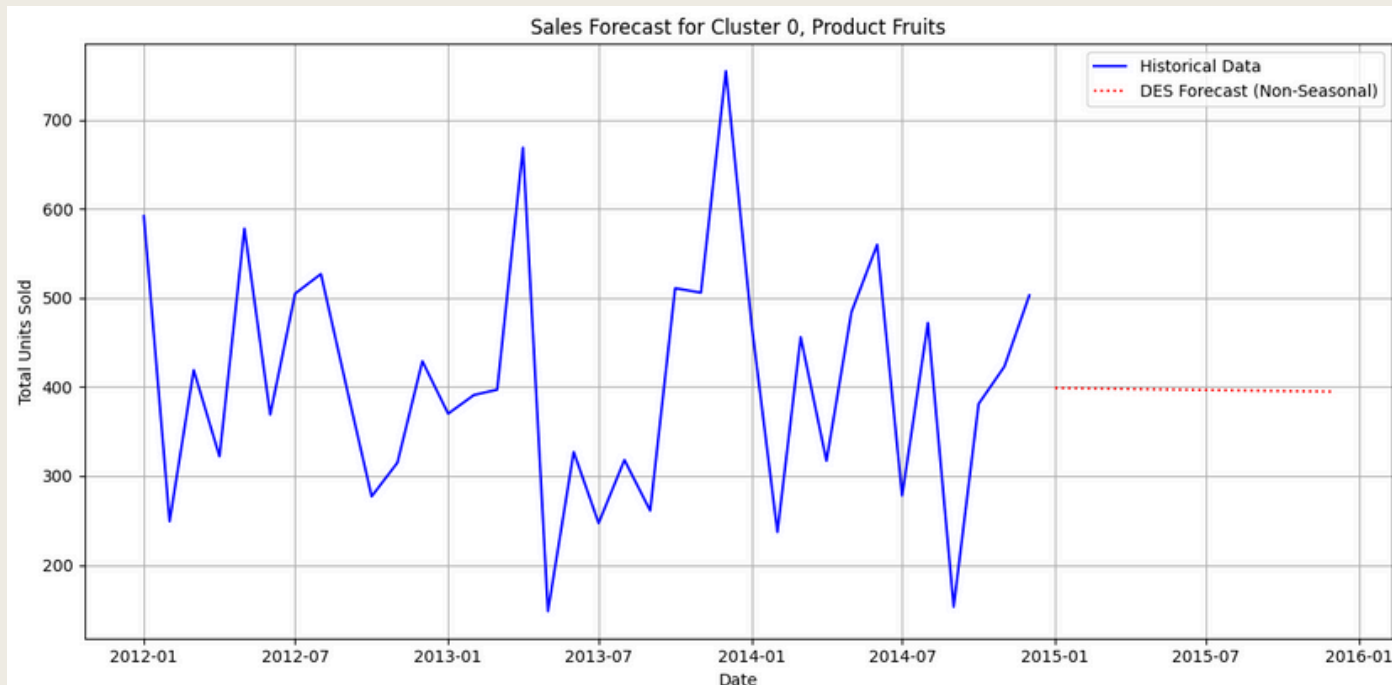
Cluster: 0, Product: Fish

Winter's Model Forecasts (2015):

2015-01-01 376.130917  
 2015-02-01 492.652451  
 2015-03-01 432.875714  
 2015-04-01 488.117810  
 2015-05-01 328.218383  
 2015-06-01 456.567980  
 2015-07-01 471.473364  
 2015-08-01 500.605000  
 2015-09-01 439.816852  
 2015-10-01 553.269650  
 2015-11-01 475.313654  
 2015-12-01 570.646030  
 Freq: MS

Double Exponential Smoothing Forecasts (2015):

2015-01-01 519.528586  
 2015-02-01 518.319265  
 2015-03-01 517.112760  
 2015-04-01 515.909063  
 2015-05-01 514.708168  
 2015-06-01 513.510068  
 2015-07-01 512.314757  
 2015-08-01 511.122228  
 2015-09-01 509.932476  
 2015-10-01 508.745492  
 2015-11-01 507.561272  
 2015-12-01 506.379808  
 Freq: MS



--- Forecast Comparison ---

Mean Winter's Model Forecast: 465.47

Mean DES Forecast: 512.93

Standard Deviation Winter's Model Forecast: 67.43

Standard Deviation DES Forecast: 4.31

Maximum difference (Winter - DES): 64.27

Minimum difference (Winter - DES): -186.49

Average Percentage Difference (Winter vs DES): -9.19%

Sample of 2015 Forecasts:

| Date          |         | 2015-01-01  | 2015-02-01  | 2015-03-01  | 2015-04-01  | \ |
|---------------|---------|-------------|-------------|-------------|-------------|---|
| Cluster_group | Product |             |             |             |             |   |
| 0             | Fish    | 376.130917  | 492.652451  | 432.875714  | 488.117810  |   |
|               | Fruits  | 456.762436  | 271.759662  | 403.425543  | 415.422147  |   |
|               | Gold    | 809.336193  | 807.998338  | 1000.999463 | 975.667509  |   |
|               | Meat    | 2467.972523 | 1782.314985 | 2282.629216 | 2325.653943 |   |
|               | Wines   | 6552.005665 | 5853.987208 | 5393.526908 | 6147.983424 |   |

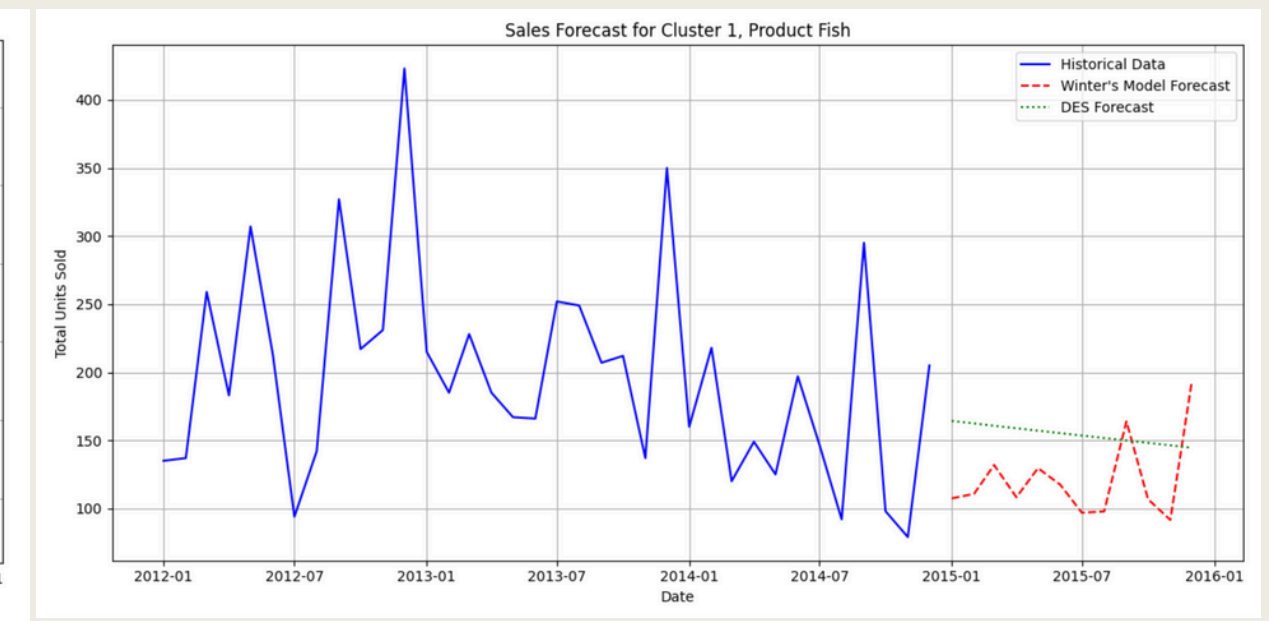
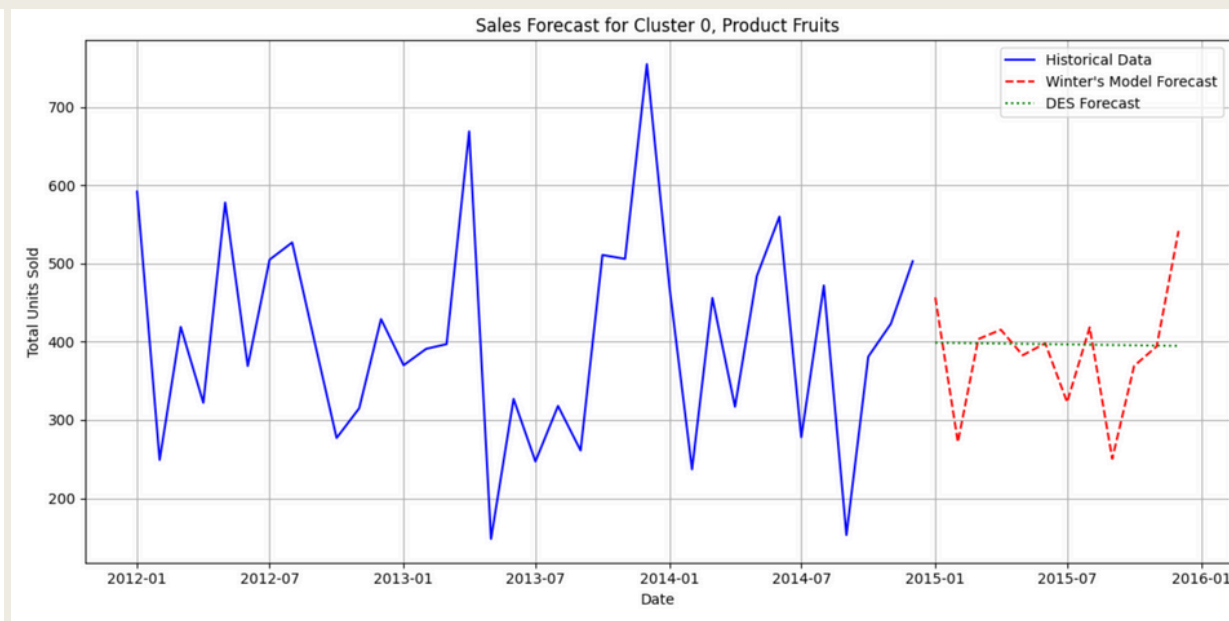
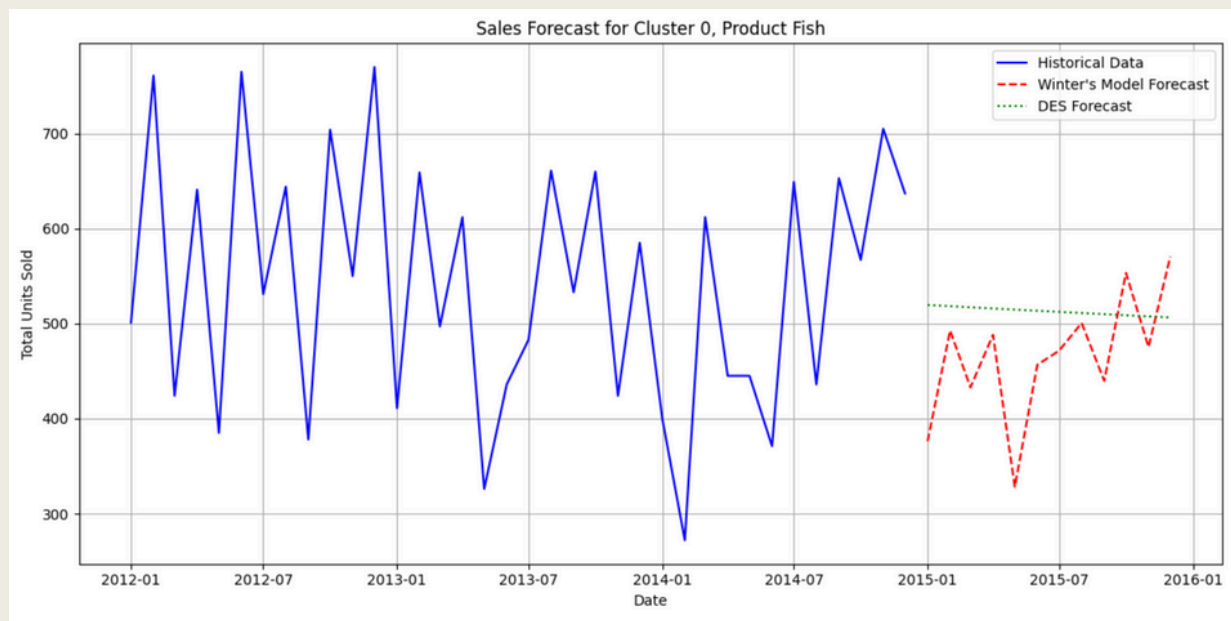
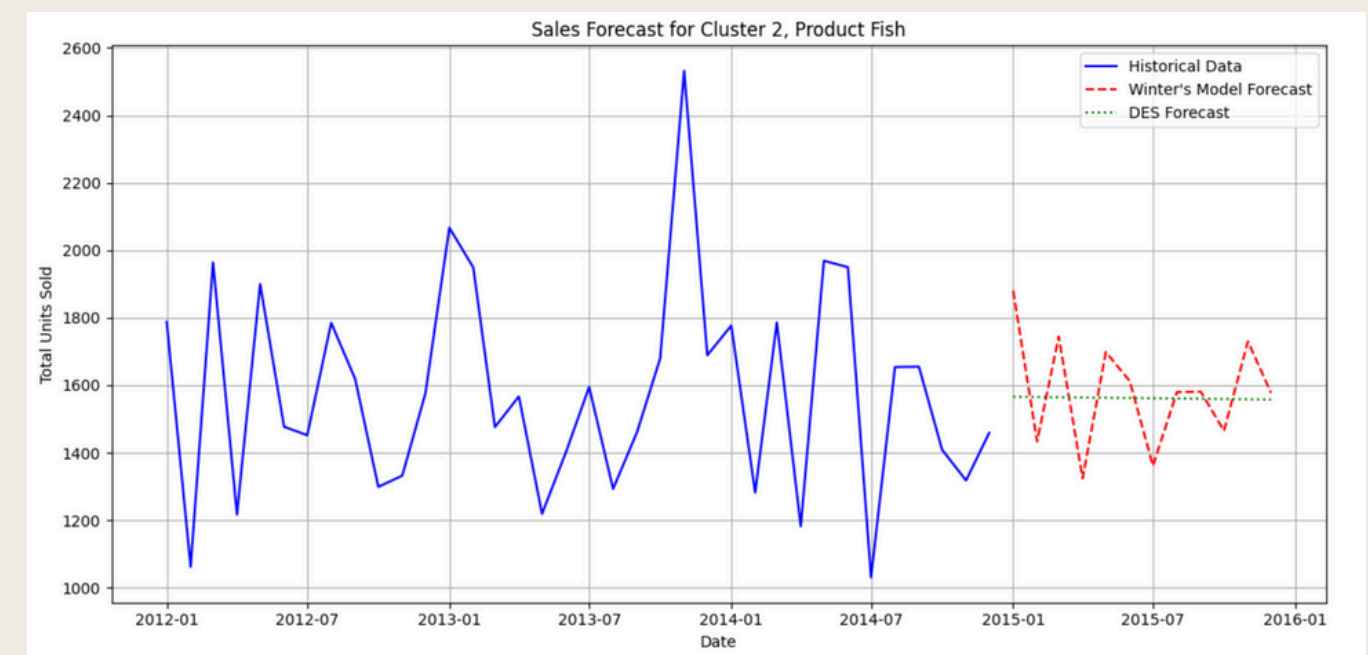
| Date          |         | 2015-05-01  | 2015-06-01  | 2015-07-01  | 2015-08-01  | \ |
|---------------|---------|-------------|-------------|-------------|-------------|---|
| Cluster_group | Product |             |             |             |             |   |
| 0             | Fish    | 328.218383  | 456.567980  | 471.473364  | 500.605000  |   |
|               | Fruits  | 382.757202  | 398.088181  | 322.754526  | 418.422783  |   |
|               | Gold    | 1007.002552 | 872.000369  | 1037.337758 | 946.332405  |   |
|               | Meat    | 2427.322816 | 1977.693052 | 1993.701326 | 2266.986679 |   |
|               | Wines   | 7874.772578 | 6427.122273 | 6531.140317 | 6195.143724 |   |

| Date          |         | 2015-09-01  | 2015-10-01  | 2015-11-01  | 2015-12-01  |  |
|---------------|---------|-------------|-------------|-------------|-------------|--|
| Cluster_group | Product |             |             |             |             |  |
| 0             | Fish    | 439.816852  | 553.269650  | 475.313654  | 570.646030  |  |
|               | Fruits  | 250.422544  | 369.091210  | 394.093017  | 541.759567  |  |
|               | Gold    | 966.000851  | 1074.665526 | 804.337823  | 700.666371  |  |
|               | Meat    | 2438.009805 | 2454.659834 | 1634.282074 | 1858.977518 |  |
|               | Wines   | 4799.729489 | 5216.348598 | 5329.395179 | 5952.900207 |  |

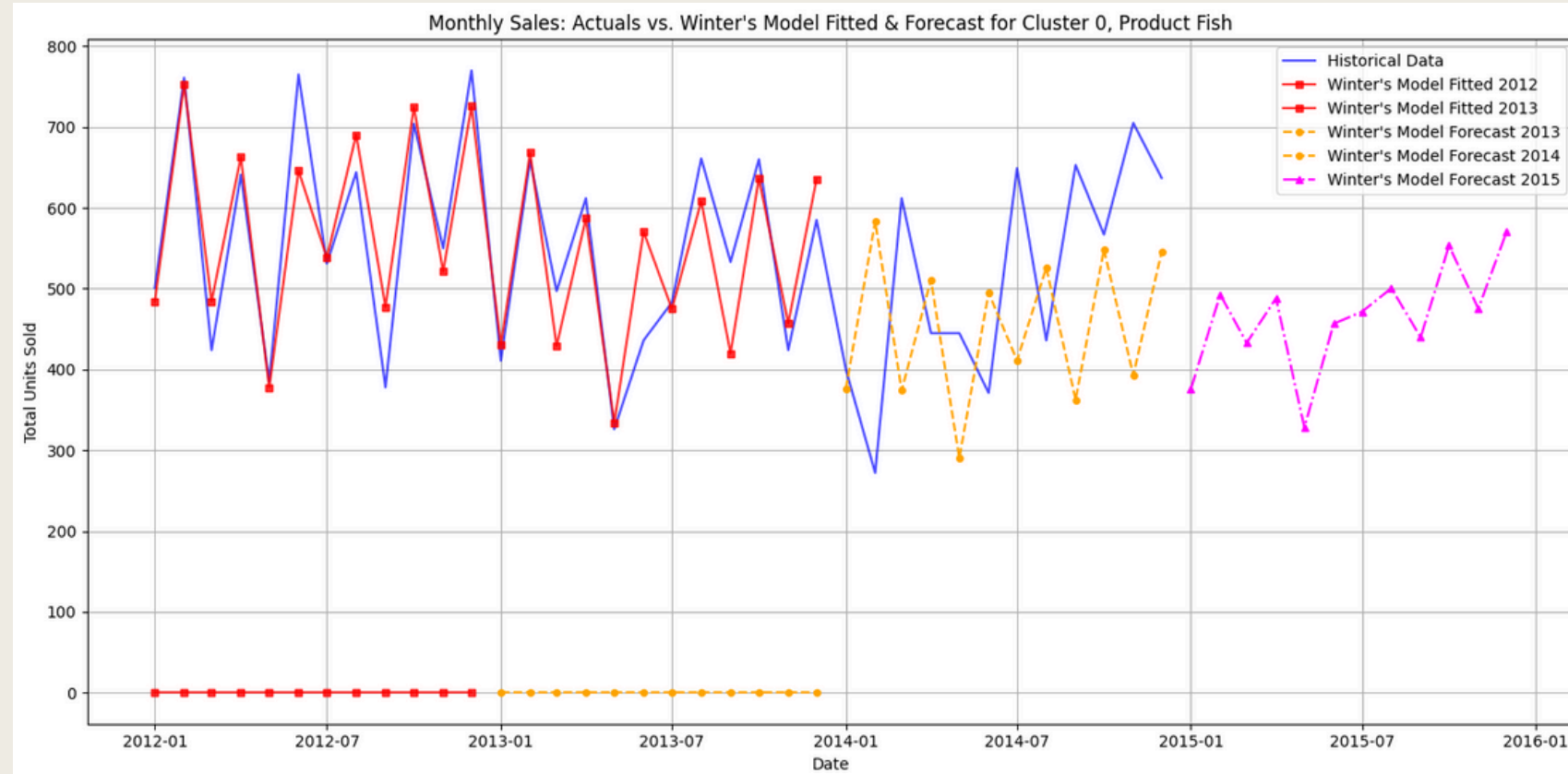
Which is  
the best  
forecast?

Winter's Model Forecast

This model considers **Seasonality**



We performed the **forecast** for **past years** to decide if the calculations were accurate



--- Overall Summary of Forecast Accuracy (2013-2014) ---  
Average MAPE (Winter's Model): 63.93%  
Average MAPE (Non-Seasonal DES): 38.94%  
Average RMSE (Winter's Model): 1474.43  
Average RMSE (Non-Seasonal DES): 604.62

General Observations:

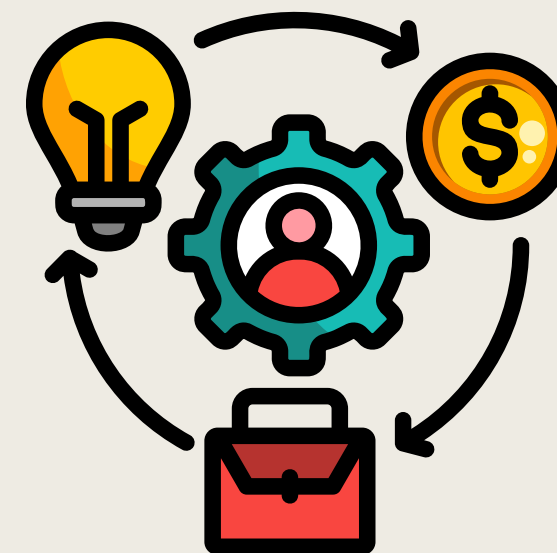
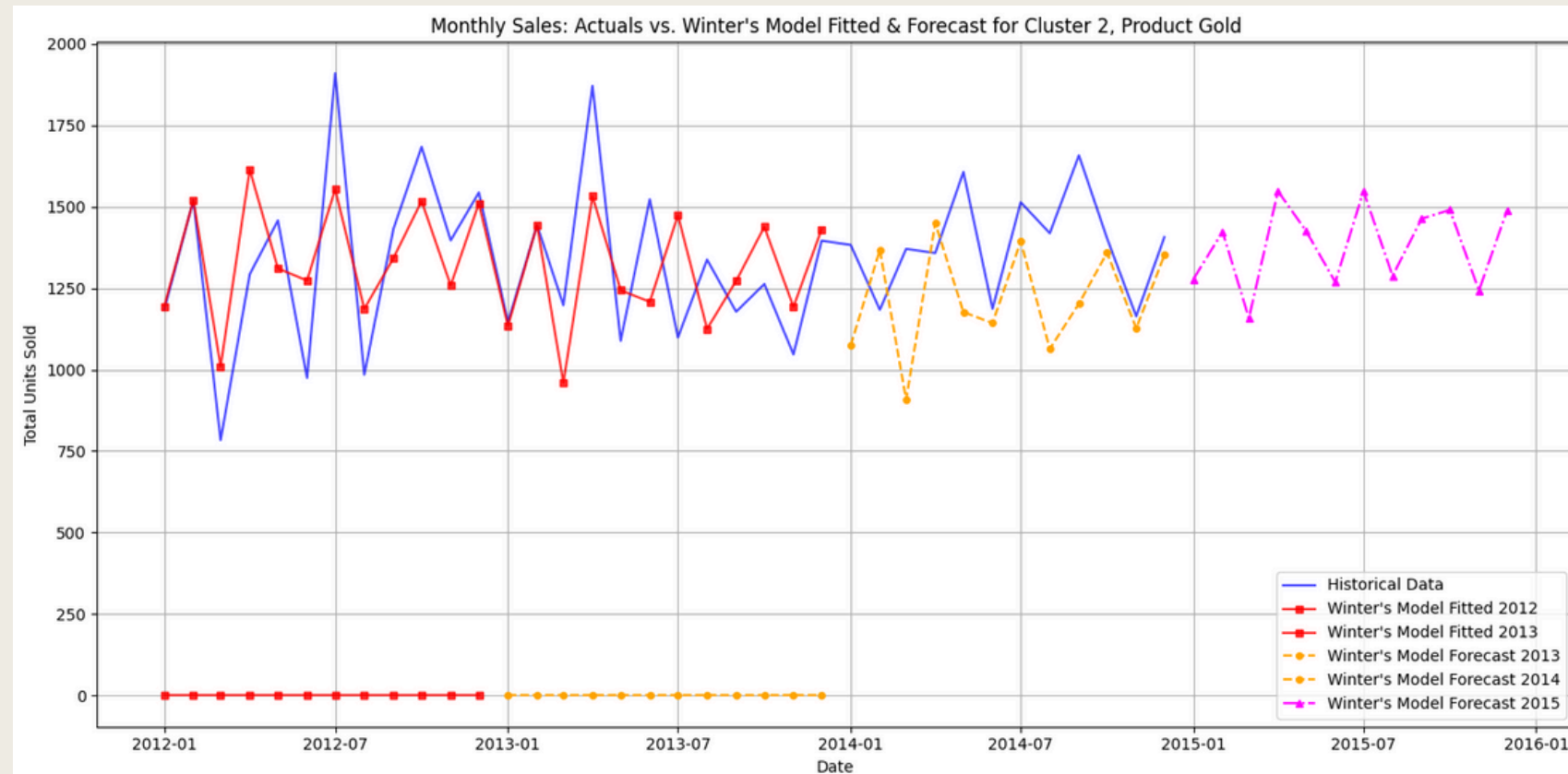
- On average, Non-Seasonal DES performed better in terms of MAPE (38.94%) than Winter's Model (63.93%).
- On average, Non-Seasonal DES performed better in terms of RMSE (604.62) than Winter's Model (1474.43).

Detailed breakdown of better-performing models by combination (MAPE):

- Winter's Model had lower MAPE for 5 combinations.
- Non-Seasonal DES had lower MAPE for 25 combinations.
- There were 0 ties in MAPE performance.

Detailed breakdown of better-performing models by combination (RMSE):

- Winter's Model had lower RMSE for 7 combinations.
- Non-Seasonal DES had lower RMSE for 23 combinations.
- There were 0 ties in RMSE performance.



While **DES** might have shown slightly better overall average **MAPE** and **RMSE** in this particular comparison for 2013-2014, **Winter's Model** is more **suitable** for sales data due to its seasonality analysis

# INVENTORY ANALYSIS

## SET

- $T = \{1, \dots, 12\} \rightarrow$  months
- Products = {Fish, Fruits, Gold, Meat, Wine}

## Parameters

- s**: set up (order) cost in period  $t$
- u**: unit price in period  $t$
- h**: holding cost in period  $t$
- i**: inventory capacity
- d**: total forecast demand in period  $t$

| Supplier information                       | Wines | Fruits | Meat   | Fish   | Gold |
|--------------------------------------------|-------|--------|--------|--------|------|
| <b>s</b> Cost per purchase (\$)            | 1500  | 500    | 550    | 550    | 2000 |
| <b>u</b> Unit Price (\$)                   | 4     | 2      | 15     | 12     | 120  |
| <b>h</b> Unit Holding Cost (\$/unit/month) | 1.25  | 1.2    | 1.2    | 1.2    | 10   |
|                                            |       |        |        |        |      |
| Store information                          | Wines | Fruits | Meat   | Fish   | Gold |
| Inventory Capacity (unit)                  | 50000 | 100000 | 100000 | 100000 | 5000 |

| month | total demand for fish | total demand for fruits | total demand for gold | total demand for meat | total demand for wine |
|-------|-----------------------|-------------------------|-----------------------|-----------------------|-----------------------|
| 1     | 2366                  | 1624                    | 2574                  | 11383                 | 19274                 |
| 2     | 2038                  | 1101                    | 2559                  | 8920                  | 16643                 |
| 3     | 2310                  | 1519                    | 2480                  | 10123                 | 18318                 |

## Variable

- q<sub>t</sub>** = quantity to buy in month  $t$
- I<sub>t</sub>** = inventory at end of month  $t$
- y<sub>t</sub>** = 1 if place order at month  $t \rightarrow$  binary

| fish | q <sub>t</sub> | I <sub>t</sub> | y <sub>t</sub> |
|------|----------------|----------------|----------------|
| 1    | 2366           | 0              | 1              |
| 2    | 2038           | 0              | 1              |
| 3    | 2310           | 0              | 1              |
| 4    | 1923           | 0              | 1              |
| 5    | 2158           | 0              | 1              |
| 6    | 2188           | 0              | 1              |
| 7    | 1931           | 0              | 1              |
| 8    | 2179           | 0              | 1              |
| 9    | 2185           | 0              | 1              |
| 10   | 2126           | 0              | 1              |
| 11   | 2299           | 0              | 1              |
| 12   | 2343           | 0              | 1              |

\*Just include the first three months (there' s no space for more)

# INVENTORY ANALYSIS

## Objective Function

$$\min Z = \sum_t (u_t q_t + s_t y_t + h_t I_t)$$

Total Cost

$$\text{Total Purchasing Cost} = \sum_t u_t q_t$$

$$\text{Total Holding Cost} = \sum_t h_t I_t$$

$$\text{Total Fixed Cost} = \sum_t s_t y_t$$

## Constraints

(1) Inventory balance:  $I_t = I_{t-1} + q_t - d_t \quad \forall t$

(2) Capacity constraint:  $I_t \leq i \quad \forall t$

(3) Order–quantity link (Big-M):  $q_t \leq M y_t \quad \forall t$   
M: sum of demand

(4) Non-negativity:  $I_t \geq 0, \quad q_t \geq 0 \quad \forall t$

(5) Binary order decision:  $y_t \in \{0, 1\} \quad \forall t$

# INVENTORY ANALYSIS

規劃求解參數

設定目標式(I)

\$B\$45

↑

至:

○ 最大值(M)

● 最小(N)

○ 值(V)

0

藉由變更變數儲存格(B)

\$C\$30:\$E\$41

↑

設定限制式(U)

\$E\$30:\$E\$41 = 二進制

\$I\$19:\$I\$30 <= \$K\$19:\$K\$30

\$I\$36:\$I\$47 <= \$K\$36:\$K\$47

\$I\$4:\$I\$15 = \$K\$4:\$K\$15

新增(A)

變更(C)

刪除(D)

全部重設(R)

載入/儲存(L)

☑ 將未設限的變數設為非負數(K)

選取求解方法(E)

單純 LP

▼

選項(P)

求解方法

針對平滑非線性的規劃求解問題，請選取 GRG 非線性引擎。針對線性規劃求解問題，請選取 LP 單純引擎，非平滑性的規劃求解問題則選取演化引擎。

說明(H)

求解(S)

關閉(O)

| set                               |                                | constrain1                                   |                               |    |        |
|-----------------------------------|--------------------------------|----------------------------------------------|-------------------------------|----|--------|
| month                             | Fish Demand (unit)             | LHS                                          | =                             |    | RHS    |
| 1                                 | 2366                           | Jan                                          | 0                             | =  | 0      |
| 2                                 | 2088                           | Feb                                          | 0                             | =  | 0      |
| 3                                 | 2310                           | March                                        | 0                             | =  | 0      |
| 4                                 | 1923                           | April                                        | 0                             | =  | 0      |
| 5                                 | 2158                           | May                                          | 0                             | =  | 0      |
| 6                                 | 2188                           | Jun                                          | 0                             | =  | 0      |
| 7                                 | 1931                           | Jul                                          | 0                             | =  | 0      |
| 8                                 | 2179                           | Aug                                          | 0                             | =  | 0      |
| 9                                 | 2185                           | Sep                                          | 0                             | =  | 0      |
| 10                                | 2126                           | Oct                                          | 0                             | =  | 0      |
| 11                                | 2299                           | Nov                                          | 0                             | =  | 0      |
| 12                                | 2343                           | Dec                                          | 0                             | =  | 0      |
| parameters                        |                                | constrain2                                   |                               |    |        |
| Supplier information              | fish                           | LHS                                          | <=                            |    | RHS    |
| Cost per purchase (\$)            | 550                            | Jan                                          | 2366                          | <= | 26046  |
| Unit Price (\$)                   | 12                             | Feb                                          | 2088                          | <= | 26046  |
| Unit Holding Cost (\$/unit/month) | 1.2                            | March                                        | 2310                          | <= | 26046  |
|                                   |                                | April                                        | 1923                          | <= | 26046  |
|                                   |                                | May                                          | 2158                          | <= | 26046  |
|                                   |                                | Jun                                          | 2188                          | <= | 26046  |
|                                   |                                | Jul                                          | 1931                          | <= | 26046  |
|                                   |                                | Aug                                          | 2179                          | <= | 26046  |
|                                   |                                | Sep                                          | 2185                          | <= | 26046  |
| Store information                 |                                | constrain3                                   |                               |    |        |
| Inventory Capacity (unit)         | Fish                           | LHS                                          | <=                            |    | RHS    |
|                                   |                                | Jan                                          | 0                             | <= | 100000 |
|                                   |                                | Feb                                          | 0                             | <= | 100000 |
|                                   |                                | March                                        | 0                             | <= | 100000 |
|                                   |                                | April                                        | 0                             | <= | 100000 |
|                                   |                                | May                                          | 0                             | <= | 100000 |
|                                   |                                | Jun                                          | 0                             | <= | 100000 |
|                                   |                                | Jul                                          | 0                             | <= | 100000 |
|                                   |                                | Aug                                          | 0                             | <= | 100000 |
|                                   |                                | Sep                                          | 0                             | <= | 100000 |
|                                   |                                | Oct                                          | 2126                          | <= | 26046  |
|                                   |                                | Nov                                          | 2299                          | <= | 26046  |
|                                   |                                | Dec                                          | 2343                          | <= | 26046  |
| variables                         |                                |                                              |                               |    |        |
|                                   | quantity to buy for each month | the inventory level at the end of each month | if u purchase or not (binary) |    |        |
| fish                              | q_t                            | I_t                                          | y_t                           |    |        |
| 1                                 | 2366                           | 0                                            | 1                             |    |        |
| 2                                 | 2088                           | 0                                            | 1                             |    |        |
| 3                                 | 2310                           | 0                                            | 1                             |    |        |
| 4                                 | 1923                           | 0                                            | 1                             |    |        |
| 5                                 | 2158                           | 0                                            | 1                             |    |        |
| 6                                 | 2188                           | 0                                            | 1                             |    |        |
| 7                                 | 1931                           | 0                                            | 1                             |    |        |
| 8                                 | 2179                           | 0                                            | 1                             |    |        |
| 9                                 | 2185                           | 0                                            | 1                             |    |        |
| 10                                | 2126                           | 0                                            | 1                             |    |        |
| 11                                | 2299                           | 0                                            | 1                             |    |        |
| 12                                | 2343                           | 0                                            | 1                             |    |        |
| TC=                               |                                | total purchasing cost                        | 312552                        |    |        |
|                                   |                                | total fixed cost                             | 6600                          |    |        |
|                                   |                                | total holding cost                           | 0                             |    |        |

fish

fruits

|           | quantity to buy for<br>each month | inventory<br>level at the<br>end of each<br>month | if u<br>purchase or<br>not (binary) |
|-----------|-----------------------------------|---------------------------------------------------|-------------------------------------|
| fruits    | q_t                               | I_t                                               | y_t                                 |
| 1         | 1624                              | 0                                                 | 1                                   |
| 2         | 1101                              | 0                                                 | 1                                   |
| 3         | 1519                              | 0                                                 | 1                                   |
| 4         | 1429                              | 0                                                 | 1                                   |
| 5         | 1447                              | 0                                                 | 1                                   |
| 6         | 1443                              | 0                                                 | 1                                   |
| 7         | 1404                              | 0                                                 | 1                                   |
| 8         | 1778                              | 0                                                 | 1                                   |
| 9         | 1565                              | 0                                                 | 1                                   |
| 10        | 1555                              | 0                                                 | 1                                   |
| 11        | 1348                              | 0                                                 | 1                                   |
| 12        | 1597                              | 0                                                 | 1                                   |
|           |                                   |                                                   |                                     |
|           |                                   |                                                   |                                     |
| TC=       | total purchasing cost             | 35620                                             |                                     |
| 41,620.00 | total fixed cost                  | 6000                                              |                                     |
|           | total holding cost                | 0                                                 |                                     |

|              | quantity to buy for<br>each month | the inventory<br>level at the<br>end of each<br>month | if u purchase<br>or not<br>(binary) |
|--------------|-----------------------------------|-------------------------------------------------------|-------------------------------------|
| Gold         | q_t                               | I_t                                                   | y_t                                 |
| 1            | 2574                              | 0                                                     | 1                                   |
| 2            | 2559                              | 0                                                     | 1                                   |
| 3            | 2480                              | 0                                                     | 1                                   |
| 4            | 2827                              | 0                                                     | 1                                   |
| 5            | 2823                              | 0                                                     | 1                                   |
| 6            | 2450                              | 0                                                     | 1                                   |
| 7            | 2842                              | 0                                                     | 1                                   |
| 8            | 2558                              | 0                                                     | 1                                   |
| 9            | 2801                              | 0                                                     | 1                                   |
| 10           | 2836                              | 0                                                     | 1                                   |
| 11           | 2401                              | 0                                                     | 1                                   |
| 12           | 2560                              | 0                                                     | 1                                   |
|              |                                   |                                                       |                                     |
|              |                                   |                                                       |                                     |
| TC=          | total purchasing cost             | 3805320                                               |                                     |
| 3,829,320.00 | total fixed cost                  | 24000                                                 |                                     |
|              | total holding cost                | 0                                                     |                                     |

gold

meat

|              | quantity to buy for<br>each month | the inventory<br>level at the<br>end of each<br>month | if u purchase<br>or not<br>(binary) |
|--------------|-----------------------------------|-------------------------------------------------------|-------------------------------------|
| Meat         | q_t                               | I_t                                                   | y_t                                 |
| 1            | 11383                             | 0                                                     | 1                                   |
| 2            | 8920                              | 0                                                     | 1                                   |
| 3            | 10123                             | 0                                                     | 1                                   |
| 4            | 10054                             | 0                                                     | 1                                   |
| 5            | 10796                             | 0                                                     | 1                                   |
| 6            | 8259                              | 0                                                     | 1                                   |
| 7            | 10233                             | 0                                                     | 1                                   |
| 8            | 11182                             | 0                                                     | 1                                   |
| 9            | 10025                             | 0                                                     | 1                                   |
| 10           | 10444                             | 0                                                     | 1                                   |
| 11           | 8863                              | 0                                                     | 1                                   |
| 12           | 9958                              | 0                                                     | 1                                   |
|              |                                   |                                                       |                                     |
|              |                                   |                                                       |                                     |
| TC=          | total purchasing cost             | 1803600                                               |                                     |
| 1 810 200,00 | total fixed cost                  | 6600                                                  |                                     |
|              | total holding cost                | 0                                                     |                                     |

|            | quantity to buy for<br>each month | the inventory<br>level at the<br>end of each<br>month | if u purchase<br>or not<br>(binary) |
|------------|-----------------------------------|-------------------------------------------------------|-------------------------------------|
| Wine       | q_t                               | I_t                                                   | y_t                                 |
| 1          | 19274                             | 0                                                     | 1                                   |
| 2          | 16643                             | 0                                                     | 1                                   |
| 3          | 18318                             | 0                                                     | 1                                   |
| 4          | 16753                             | 0                                                     | 1                                   |
| 5          | 20116                             | 0                                                     | 1                                   |
| 6          | 17455                             | 0                                                     | 1                                   |
| 7          | 17493                             | 0                                                     | 1                                   |
| 8          | 18515                             | 0                                                     | 1                                   |
| 9          | 17987                             | 0                                                     | 1                                   |
| 10         | 18619                             | 0                                                     | 1                                   |
| 11         | 19594                             | 0                                                     | 1                                   |
| 12         | 18836                             | 0                                                     | 1                                   |
|            |                                   |                                                       |                                     |
|            |                                   |                                                       |                                     |
| TC=        | total purchasing cost             | 878412                                                |                                     |
| 896 412,00 | total fixed cost                  | 18000                                                 |                                     |
|            | total holding cost                | 0                                                     |                                     |

wine

# CONCLUSION

The optimization results show that ending inventory is zero every month across all five product categories Fish Fruits, Meat, Gold, and Wine.

- Demand is met precisely each month without leftover stock
- Replenishment timing can be highly accurate



Managerial Recommendation:  
Adopt Just-In-Time (JIT)  
replenishment system to  
reduce holding costs and  
improve inventory efficiency.

**THANK  
YOU!**



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